

TSA Complaints Analysis

Audience:

Understanding passenger complaints is an essential part of improving airport security operations and maintaining trust in the Transportation Security Administration (TSA). The target audience for this project is the TSA operations leadership and airport management teams. These individuals are familiar with airport procedures and security operations, but don't handle airport complaint analytics. The data used for this analysis comes from TSA complaint datasets, which include a code lookup file (IATA/ICAO). This information provides insight into where and why passengers file complaints, allowing patterns and problem areas to be identified across different airports.

The audience consists of internal stakeholders with moderate data knowledge. The story will use clear visuals and simple language to communicate findings. The visuals will highlight key patterns, such as airports with the highest complaint rates or the most common categories of complaints. The goal is to help TSA decision-makers easily interpret the data, identify areas needing improvement, and take action to enhance passenger satisfaction and operational efficiency.

Purpose:

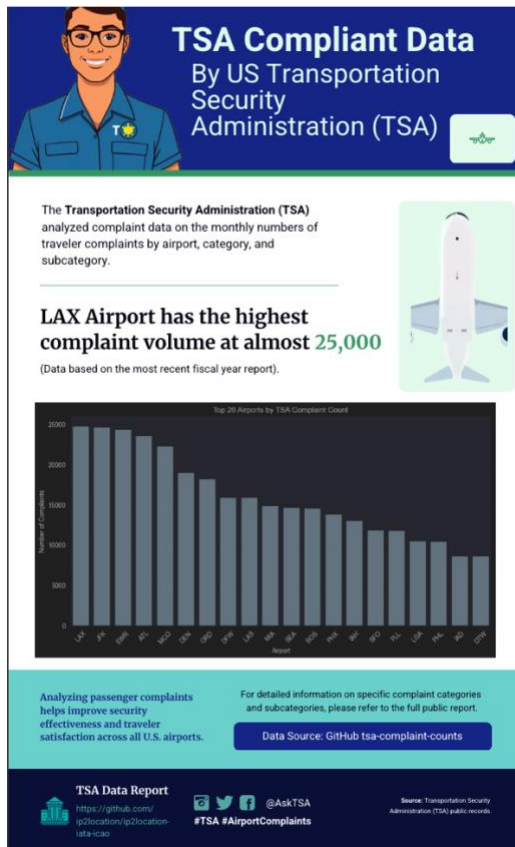
The purpose of this project is to use TSA complaint data to identify key areas where passengers experience issues and to recommend specific actions that can improve service and efficiency. By evaluating complaints by airport, category, and subcategory, the goal is to uncover

trends that highlight which airports or types of issues occur most frequently. This analysis will not only help the TSA leadership understand where issues are the most common, but also provide data-based insights for improving security screening processes, communication, and staffing strategies.

Call to Action:

The call to action is to encourage TSA decision-makers to use the data to implement targeted improvements at the airports with the highest complaint rates. By thoroughly analyzing the data, the TSA can address common traveler concerns, enhance passenger satisfaction, and strengthen confidence in the agency's operation.

Infographic:



In the infographic, I used the Graph 4: Boxplot - Top 20 Airports by Complaint Count, created on Python. The infographic was created as a poster on Venngage to ideally use at the beginning of a presentation.

Medium for Infographic:

The information will be presented as a poster-style infographic created with Venngage, along with a graph generated in Python. This format is effective and clean for TSA operations leadership and airport management because it highlights the top twenty airports with the highest number of complaints. It clearly identifies the airports that need immediate attention, with LAX, JFK, and EWR being the top three. Allowing decision-makers to focus on areas where improvements can have the greatest impact.

Design for Infographic:

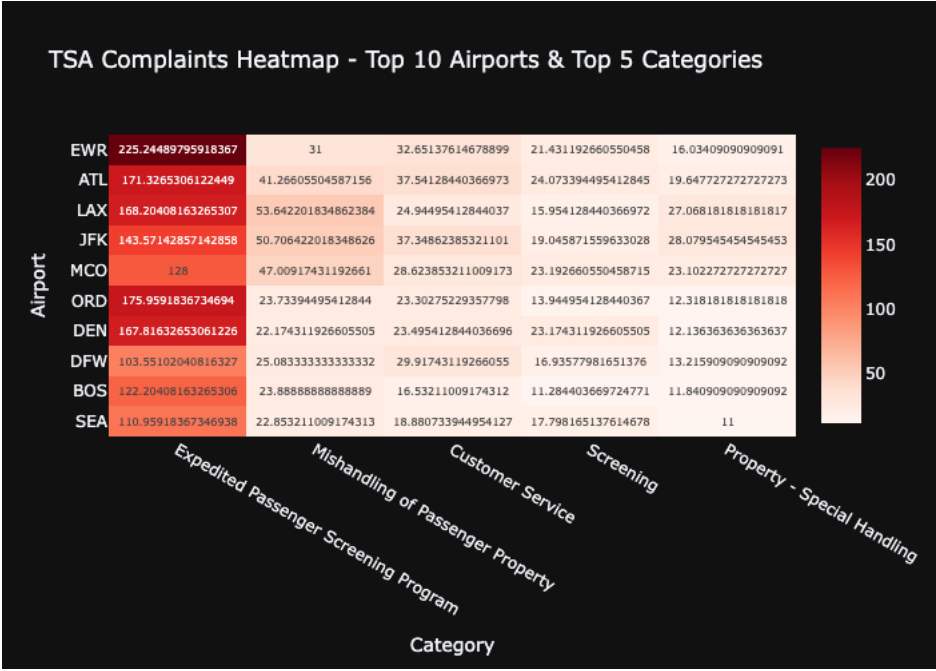
The design of the poster uses clear visual organization to make the data easy to understand. Gestalt principles guided the design to make the infographic intuitive and easy to read (Cherry, 2024). Focusing on a blue color scheme, with a variety of white and blue shades. Then there is a graph made from Python. Along with a few little icons/pictures to keep the user engaged, like a picture of a friendly man and a small, cute plane. Since this is designed to be a handout at the beginning of a presentation, there is only 3 headlines but not a lot of factual information about the whole project. Color, text, alignment, sizing, and spacing were carefully chosen to improve readability and guide the audience's attention to the most important insights.

Visuals / Graphs from Python:

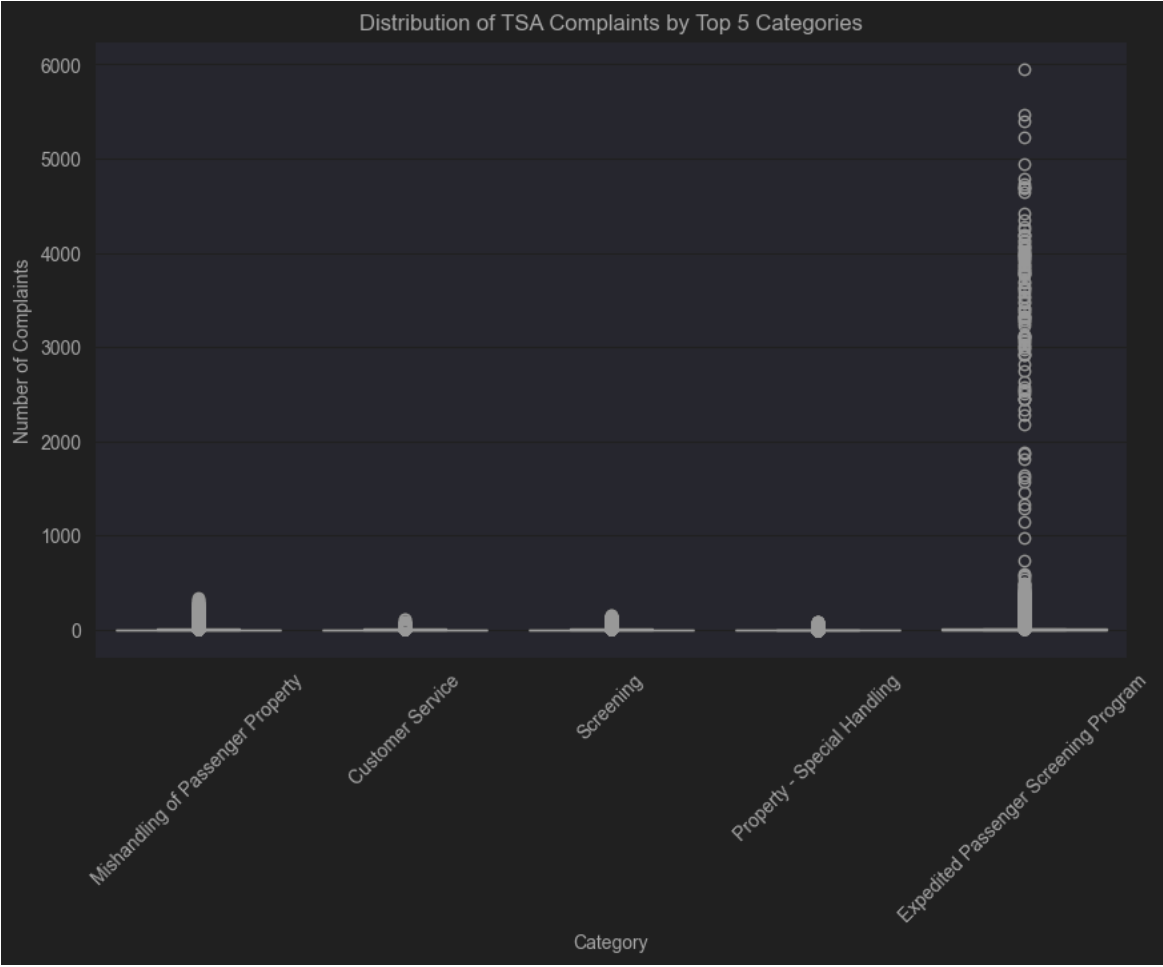
Graph 1: Spatial / Choropleth Map



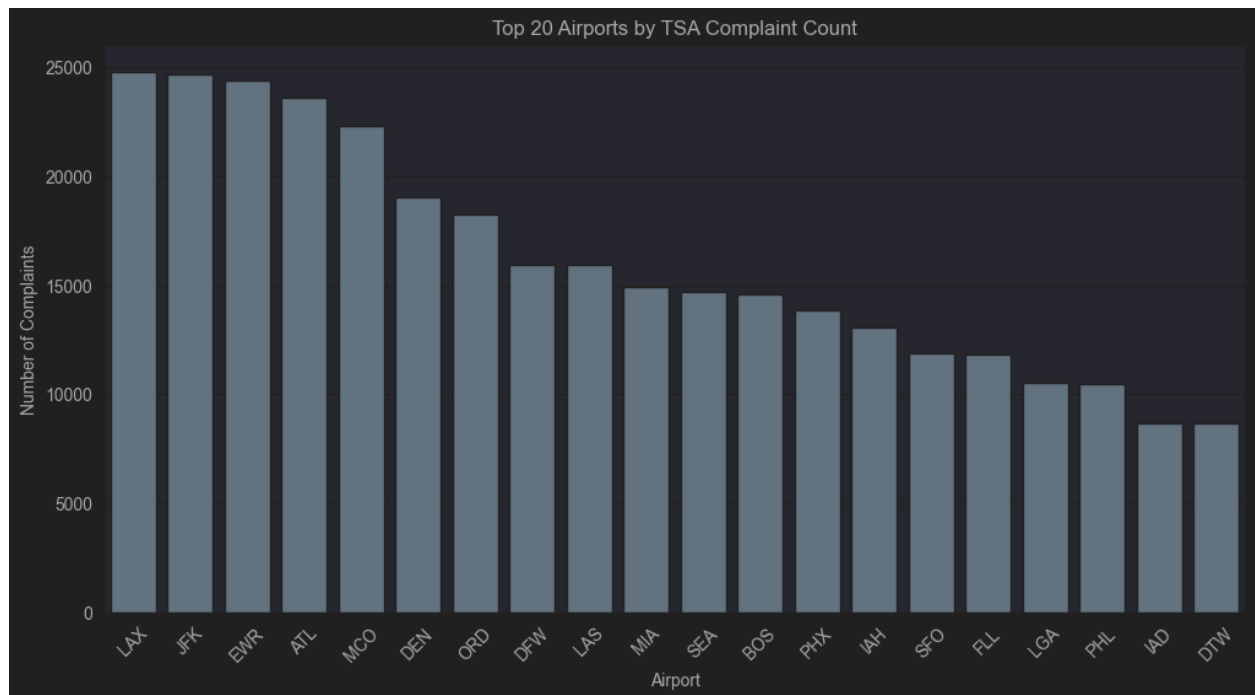
Graph 2: Heat Map - Airports & Categories



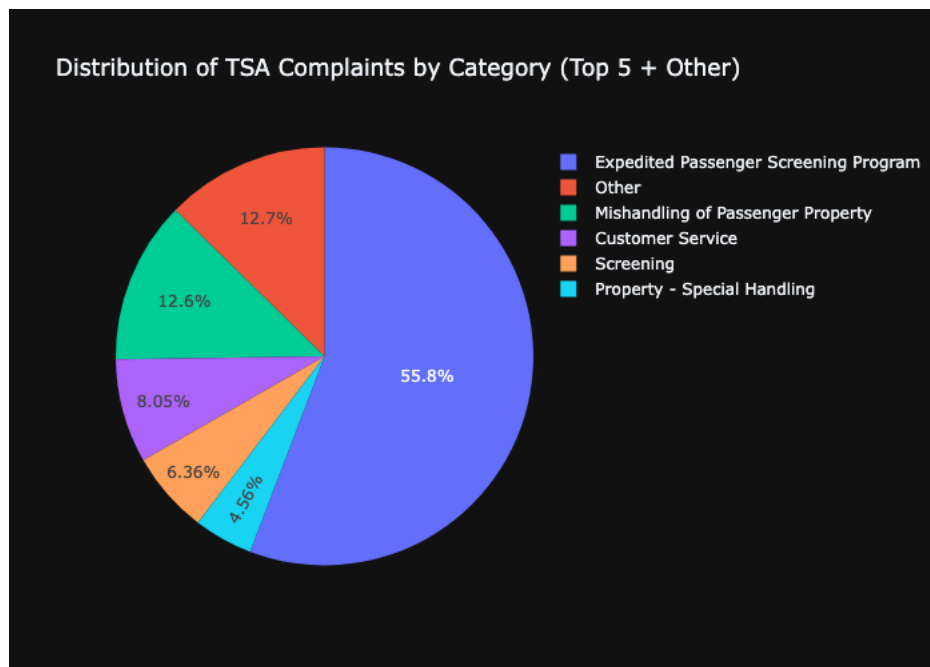
Graph 3: Boxplot - Complaints by Category



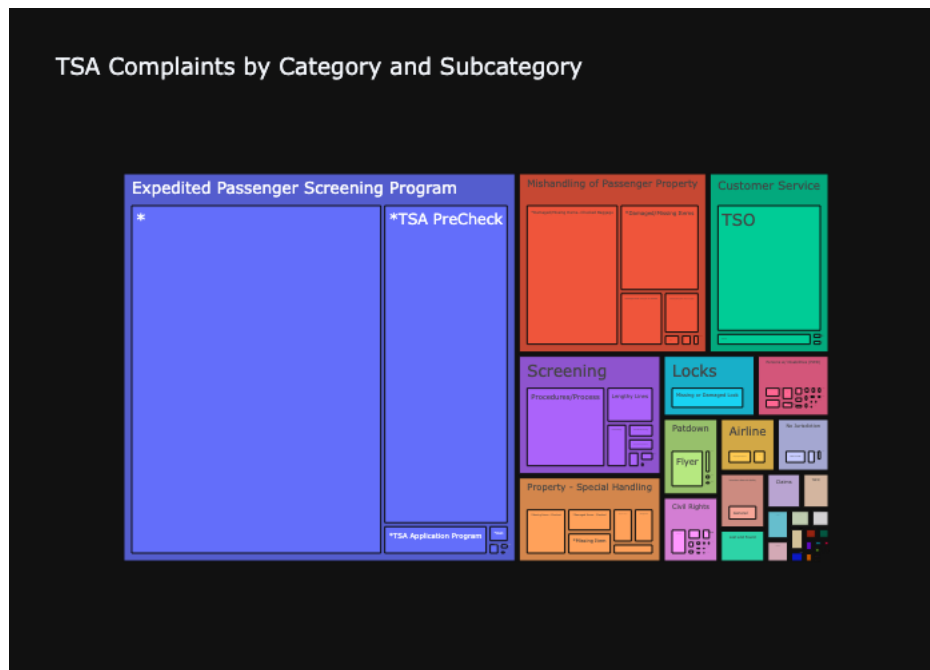
Graph 4: Boxplot - Top 20 Airports by Complaint Count



Graph 5: Pie Chart - Complaints Distribution by Category



Graph 6: Treemap - Subcategory Complaints



All 6 graphs were made with Python.

Ethical Considerations:

When analyzing the TSA complaint data, steps were taken to protect passenger privacy and ensure transparency in the changes made. The data was separated by airport, category, and subcategory to focus on trends rather than individual complaints. Assumptions and transformations, such as mapping missing airport codes or normalizing complaints by passenger volume, were clearly documented to avoid misrepresentation of the datasets.

Conclusion:

This analysis of the TSA complaint data shows where passengers have the most complaints at which airports and what types of issues are most common. The Python graphs and infographic make the information easier to understand for TSA leadership and airport managers. By focusing on the top airports, like LAX, JFK, and EWR, the TSA can take action to improve

passenger experience, on which airports need it the most. Overall, the project helps the TSA make better decisions to fix problems and make security processes more efficient.

Resources:

PyCharm – Python – Jupyter Notebook

[Venngage](#)

Cherry, K. (2024) *What Are the Gestalt Principles?*

<https://www.verywellmind.com/gestalt-laws-of-perceptual-organization-2795835>

Datasets:

[Complaints by Airport](#)

[Complaints by Category](#)

[Complaints by Subcategory](#)

[IATA/ICAO](#)

Data Source:

[GitHub tsa-complaint-counts](#)

[IATA/ICAO](#)